

Questions with Answer Keys

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Q1 - HP

Let $a_1, a_2, \dots, a_{10}$ be in AP and $h_1, h_2, \dots, h_{10}$ be in HP. If $a_1 = h_1 = 2$ and $a_{10} = h_{10} = 3$, then $a_4 h_7$ is

- (1) 2
- (2) 3
- (3) 5
- (4) 6

Q2 - HP

The sixth HM between 3 and $\frac{6}{13}$ is

- (1) $\frac{63}{120}$
- (2) $\frac{63}{12}$
- (3) $\frac{126}{105}$
- (4) $\frac{120}{63}$

Q3 - HP

If $b + c, c + a, a + b$ are in HP, then $\frac{a}{b+c}, \frac{b}{c+a}, \frac{c}{a+b}$ are in

- (1) AP
- (2) HP
- (3) GP
- (4) None of these

Q4 - HP

If $a_1, a_2, \dots, a_n$ are in HP, then the expression $a_1 a_2 + a_2 a_3 + \dots + a_{n-1} a_n$ is equal to

- (1) $(n-1)(a_1 - a_n)$
- (2) $na_1 a_n$
- (3) $(n-1)a_1 a_n$
- (4) $n(a_1 - a_n)$

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Q5 - HP

If the $(m+1)$ th, $(n+1)$ th and $(r+1)$ th terms of an AP are in GP and m, n, r are in HP. Then, the value of the ratio of the common difference to the first term of the AP is

(1) $-\frac{2}{n}$

(2) $\frac{2}{n}$

(3) $-\frac{n}{2}$

(4) $\frac{n}{2}$

Q6 - Special Series

The sum of infinite terms of the series $1 + \frac{4}{5} + \frac{7}{5^2} + \frac{10}{5^3} + \dots$ will be

(1) $\frac{3}{16}$

(2) $\frac{35}{8}$

(3) $\frac{35}{4}$

(4) $\frac{35}{16}$

Q7 - Special Series

$\sum_{i=1}^n \sum_{j=1}^i \sum_{k=1}^j 1$, is equal to

(1) $\frac{n(n+1)(2n+1)}{6}$

(2) $\left(\frac{n(n+1)}{2}\right)^2$

(3) $\frac{n(n+1)}{2}$

(4) $\frac{n(n+1)(n+2)}{6}$

Q8 - Special Series

If $\frac{1}{1^4} + \frac{1}{2^4} + \frac{1}{3^4} + \dots \infty = \frac{\pi^4}{90}$, then the value of $\frac{1}{1^4} + \frac{1}{3^4} + \frac{1}{5^4} + \dots \infty$ is

(1) $\frac{\pi^4}{96}$

(2) $\frac{\pi^4}{45}$

(3) $\frac{89}{90} \pi^4$

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(4) None of these

Q9 - Special Series

$1 + 2 \cdot 2 + 3 \cdot 2^2 + 4 \cdot 2^3 + \dots + 100 \cdot 2^{99}$ is equal to

(1) $99 \cdot 2^{100}$

(2) $100 \cdot 2^{100}$

(3) $1 + 99 \cdot 2^{100}$

(4) None of these

Q10 - Special Series

The sum of $1 + (1 + x) + (1 + x + x^2) + (1 + x + x^2 + x^3) + \dots$ upto n terms, equals

(1) $\frac{1-x^n}{1-x}$

(2) $\frac{x(1-x^n)}{1-x}$

(3) $\frac{n(1-x) - x(1-x^n)}{(1-x)^2}$

(4) None of these

Q11 - Special Series

The sum of $1 + 2\left(1 + \frac{1}{n}\right) + 3\left(1 + \frac{1}{n}\right)^2 + 4\left(1 + \frac{1}{n}\right)^3 + \dots$ upto n terms, equals

(1) $n\left(1 + \frac{1}{n}\right)$

(2) n^2

(3) $n\left(1 + \frac{1}{n}\right)^2$

(4) None of these

Q12 - Special Series

The value of $\sum_{n=1}^{\infty} (-1)^{n+1} \left(\frac{n}{5^n}\right)$ equals

(1) $\frac{5}{12}$

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(2) $\frac{5}{24}$

(3) $\frac{5}{36}$

(4) $\frac{5}{16}$

Q13 - Special Series

The sum to infinity of the series $\frac{1}{1} + \frac{1}{1+2} + \frac{1}{1+2+3} + \dots$ is equal to

(1) 2

(2) $\frac{5}{2}$

(3) 3

(4) None of these

Q14 - Special Series

If $abcd = 1$, where a, b, c, d are positive reals, then the minimum value of

$a^2 + b^2 + c^2 + d^2 + ab + ac + ad + bc + bd + cd$ is

(1) 6

(2) 10

(3) 12

(4) 20

Q15 - Special Series

The sum of $\sum_{k=1}^{100} \frac{k}{k^4 + k^2 + 1}$ is equal to

(1) $\frac{4950}{10101}$

(2) $\frac{5050}{10101}$

(3) $\frac{5151}{10101}$

(4) None of these

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Answer Key

Q1 (4)	Q2 (1)	Q3 (1)	Q4 (3)
Q5 (1)	Q6 (4)	Q7 (4)	Q8 (1)
Q9 (3)	Q10 (3)	Q11 (2)	Q12 (3)
Q13 (1)	Q14 (2)	Q15 (2)	